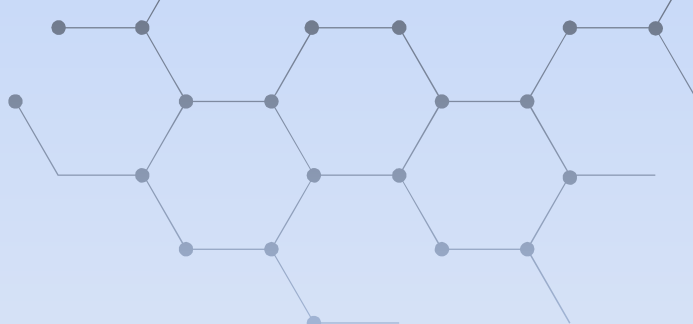
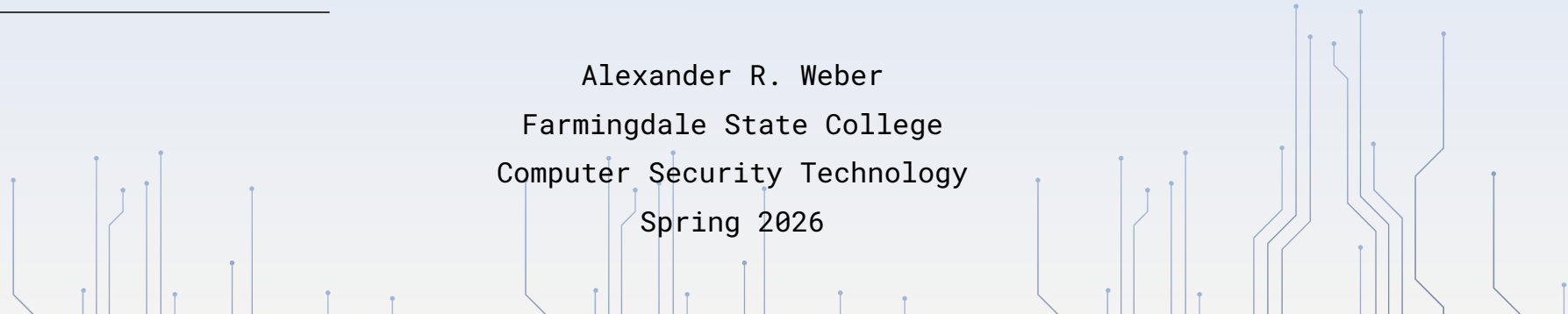
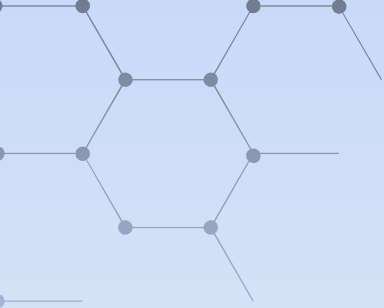




Deploying Cybersecurity Honeypots

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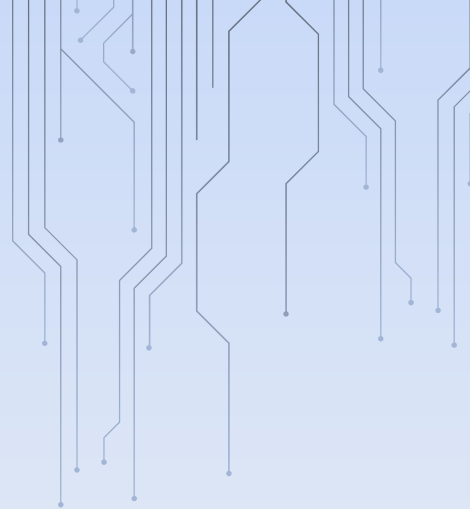
Overview

- Cyber threats are increasing
- Traditional defenses have limitations
- Honeypots attract attackers
- AWS + T-Pot deployment
- Goal: analyze attacker behavior



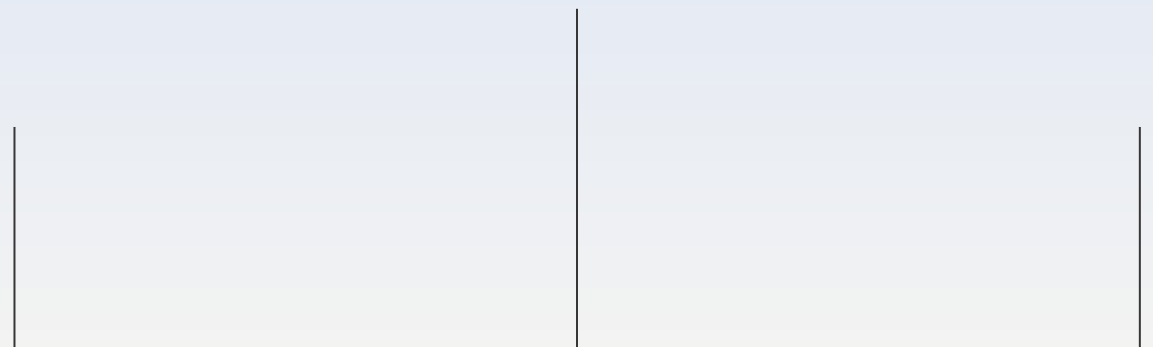
Introduction

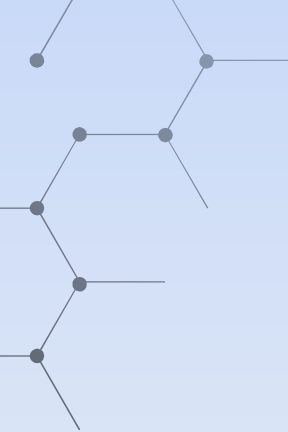
- Cyber threats are evolving
- Attackers exploit vulnerabilities
- Traditional tools miss behavior insights
- Honeypots capture real interactions





Research Purpose

- Evaluate honeypot effectiveness
 - Capture real attack data
 - Identify attack patterns
 - Bridge theory and practice
- 



Literature Review

- Honeypots provide real-time insights
 - Generate high-quality data
 - Capture scanning & brute-force attacks
 - Support digital forensics
- 

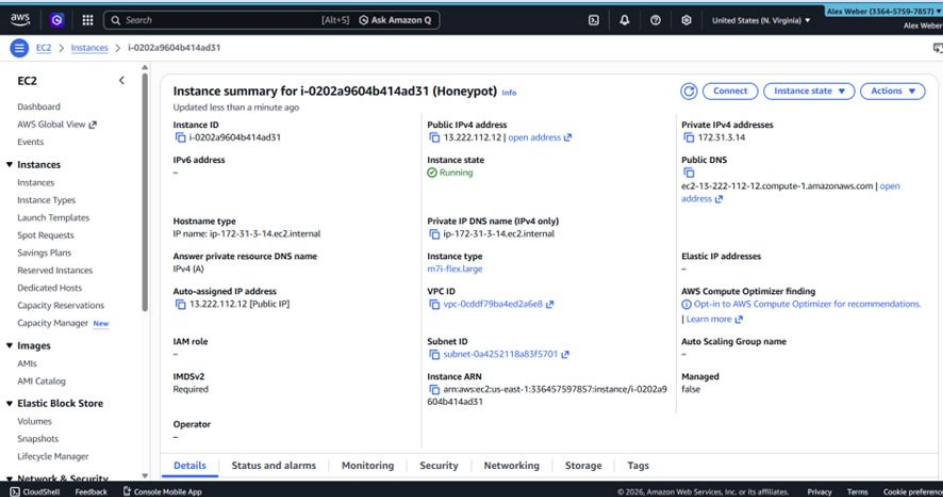
Methodology



- Controlled honeypot environment
 - Simulated vulnerabilities
 - Cloud-based deployment
 - Logs used for analysis
- 

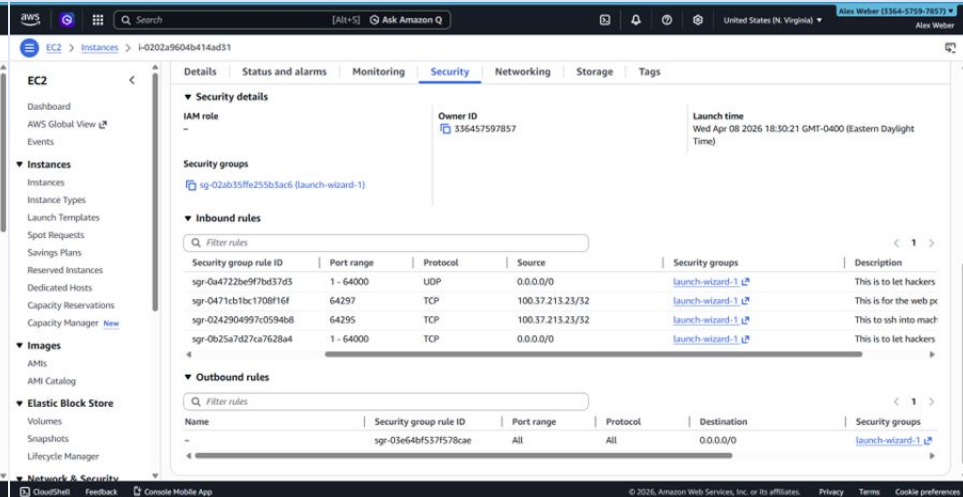
Deployment Setup

- AWS EC2 instance
- T-Pot platform
- Ubuntu Linux
- Public IP enabled



The screenshot displays the AWS Management Console for an EC2 instance named 'i-0202a9604b414ad31' (HoneyPot). The instance is in a 'Running' state. Key details include:

- Instance ID:** i-0202a9604b414ad31
- Public IPv4 address:** 13.222.112.12
- Private IPv4 addresses:** 172.31.3.14
- Public DNS:** ec2-13-222-112-12.compute-1.amazonaws.com
- Private DNS name (IPv4 only):** ip-172-31-3-14.ec2.internal
- Instance type:** m7i-flex.large
- VPC ID:** vpc-0cd0f79ba4ed2a6e8
- Subnet ID:** subnet-0a4252118a83f5701
- Instance ARN:** arn:aws:ec2:us-east-1:336457597857:instance/i-0202a9604b414ad31
- IAM role:** AmazonEC2RoleforAWSLambda
- Operator:** AmazonEC2RoleforAWSLambda



The screenshot displays the 'Security' tab for the instance 'i-0202a9604b414ad31'. It shows the following details:

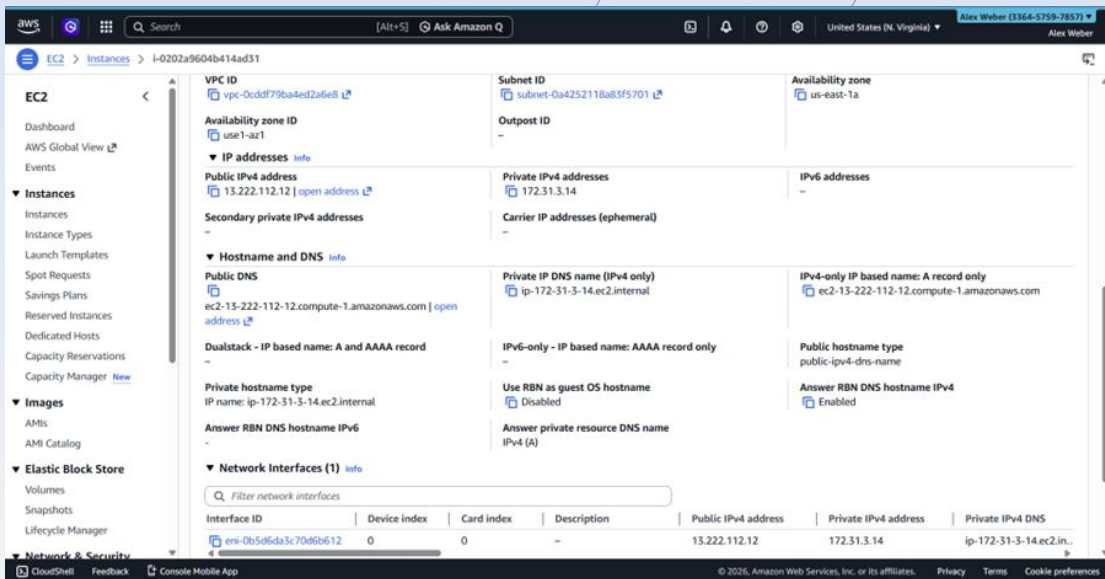
- Security group:** sg-02ab35f0e255b3ac6 (launch-wizard-1)
- Inbound rules:** A table listing three rules for the security group.
- Outbound rules:** A table listing one rule for the security group.

Security group rule ID	Port range	Protocol	Source	Security groups	Description
sgr-0a4722be9f7bd37d3	1 - 64000	UDP	0.0.0.0/0	launch-wizard-1	This is to let hackers
sgr-0471cb10c170816f	64297	TCP	100.37.213.23/32	launch-wizard-1	This is for the web pc
sgr-0242904997c0594b8	64295	TCP	100.37.213.23/32	launch-wizard-1	This is to sh into mach

Name	Security group rule ID	Port range	Protocol	Destination	Security groups
-	sgr-03e64bf537f578cae	All	All	0.0.0.0/0	launch-wizard-1

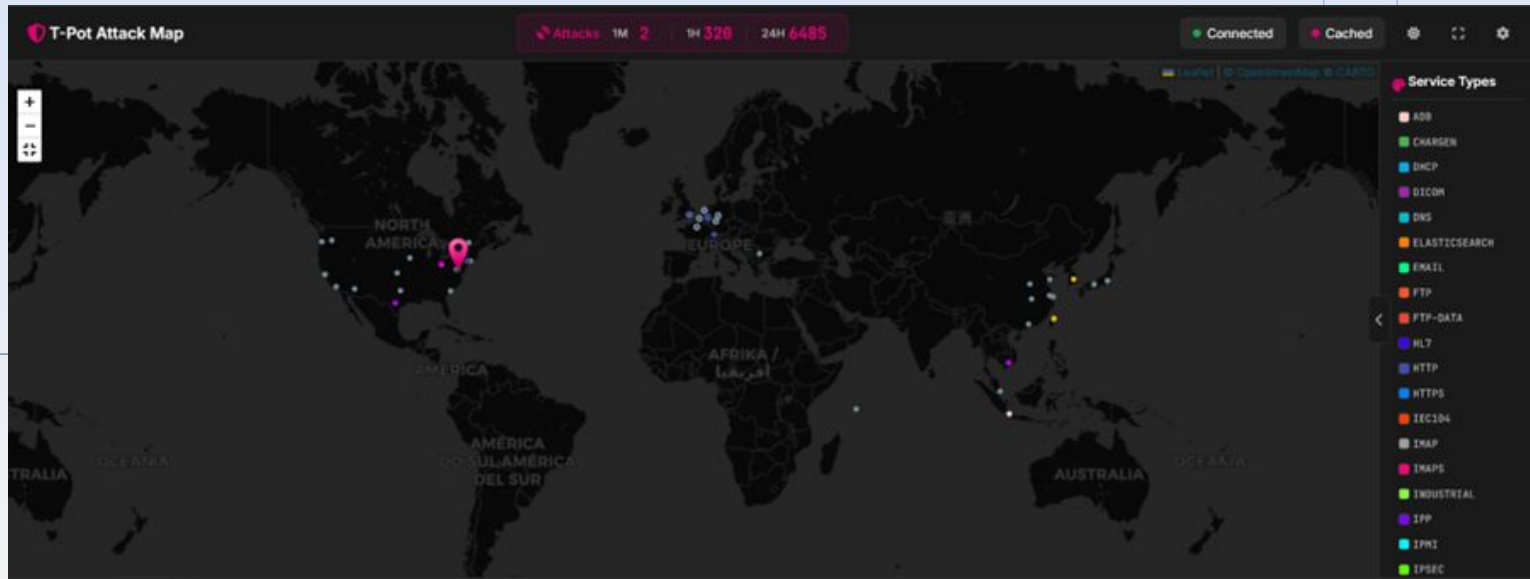
Architecture Design

- Internet-facing honeypot
- Exposed services
- Logging components
- Isolated environment



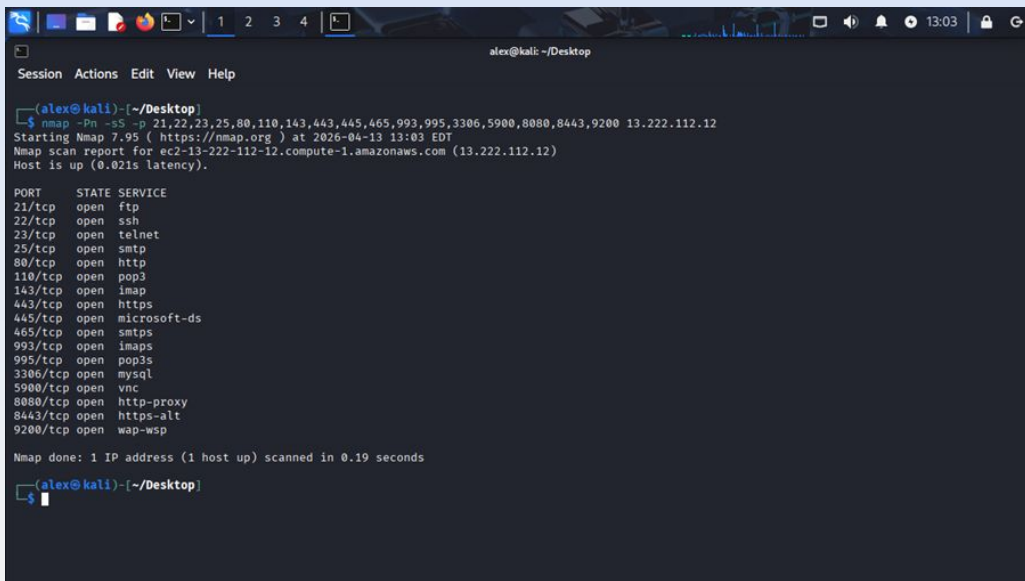
Data Collection

- Cowrie (SSH/Telnet)
- Dionaea (network attacks)
- Herdling (auth attempts)
- Logs: IPs, timestamps, commands



Attack Simulation

- Nmap scanning
- Hydra brute force
- Web probing tools
- Generated realistic attack data



The screenshot shows a Kali Linux terminal window with the following content:

```
alex@kali: ~/Desktop
Session Actions Edit View Help

(alex@kali) [~/Desktop]
$ nmap -Pn -ss -p 21,22,23,25,80,110,143,443,445,465,993,995,3306,5900,8080,8443,9200 13.222.112.12
Starting Nmap 7.95 ( https://nmap.org ) at 2026-04-13 13:03 EDT
Nmap scan report for ec2-13-222-112-12.compute-1.amazonaws.com (13.222.112.12)
Host is up (0.021s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
110/tcp   open  pop3
143/tcp   open  imap
443/tcp   open  https
445/tcp   open  microsoft-ds
465/tcp   open  smtps
993/tcp   open  imaps
995/tcp   open  pop3s
3306/tcp  open  mysql
5900/tcp  open  vnc
8080/tcp  open  http-proxy
8443/tcp  open  https-alt
9200/tcp  open  wap-wsp

Nmap done: 1 IP address (1 host up) scanned in 0.19 seconds

(alex@kali) [~/Desktop]
$
```

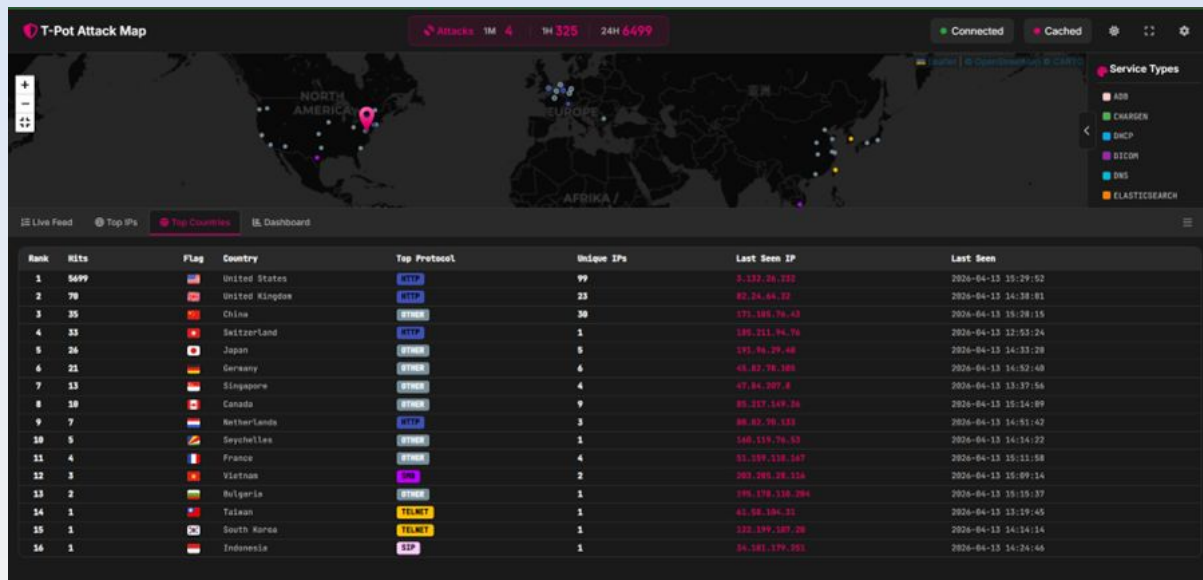
[illegible]

-

[illegible]

Attack Trends

- Global attack sources
- HTTP most targeted
- SSH, SMB, Telnet also targeted
- Automated distributed attacks



Discussion

- High-quality data collection
- Early threat detection
- Useful in small-scale environments
- Insight into attacker strategies

Session	Actions	Edit	View	Help
admin@ip-172-31-3-14:~\$	docker ps			
CONTAINER ID	NAME	COMMAND	CREATED	STATUS
8966376d9d1	gchr.io/telium-security/mare:24.04.1	"/bin/sh -c 'share -u'"	41 minutes ago	Up 41 minutes
626490014dc	gchr.io/telium-security/tanner:24.04.1	"tanner"	41 minutes ago	Up 41 minutes
399d714d073	gchr.io/telium-security/kthana:24.04.1	"docker-entrypoint.sh"	41 minutes ago	Up 39 minutes (healthy)
fcd45f8f7db	gchr.io/telium-security/tegitash:24.04.1	"entrypoint.sh"	41 minutes ago	Up 39 minutes (healthy)
21f8584f028	gchr.io/telium-security/nap:24.04.1	"/bin/sh -c './usr/bin/..."	41 minutes ago	Up 39 minutes
a7336362080	gchr.io/telium-security/tanner:24.04.1	"tannerapi"	41 minutes ago	Up 41 minutes
a07614900f6	gchr.io/telium-security/honeyapi:24.04.1	"/homeapi -d /opt/..."	41 minutes ago	Up 41 minutes
a21f6338cc	gchr.io/telium-security/healdog:24.04.1	"/bin/sh -c 'exec sh'"	41 minutes ago	Up 41 minutes
1111445746b	gchr.io/telium-security/healdog:24.04.1	"/bin/sh -c 'exec sh'"	41 minutes ago	Up 41 minutes
fcd6c88f485	gchr.io/telium-security/mailbox:24.04.1	"/usr/bin/python mail..."	41 minutes ago	Up 41 minutes
780c123577	gchr.io/telium-security/spiderfoot:24.04.1	"/bin/sh -c 'echo -u'"	41 minutes ago	Up 41 minutes (healthy)
6329a52271	gchr.io/telium-security/compet:24.04.1	"/bin/sh -c 'exec ./u..."	41 minutes ago	Up 41 minutes (healthy)
79f75056465	gchr.io/telium-security/elastiprot:24.04.1	"/elastiprot"	41 minutes ago	Up 41 minutes
306279d6c7	gchr.io/telium-security/wodpost:24.04.1	"/wodpost --host 0.0.0.0..."	41 minutes ago	Up 41 minutes
720726206d	gchr.io/telium-security/lphoney:24.04.1	"/lphoney"	41 minutes ago	Up 41 minutes
4a6d024721	gchr.io/telium-security/entrypoint:24.04.1	"/bin/sh -c './usr/bin/..."	41 minutes ago	Up 41 minutes
716d350842	gchr.io/telium-security/ciscosca:24.04.1	"/bin/sh -c './asa.c..."	41 minutes ago	Up 41 minutes
8a14370623a	gchr.io/telium-security/suricata:24.04.1	"entrypoint.sh"	41 minutes ago	Up 41 minutes

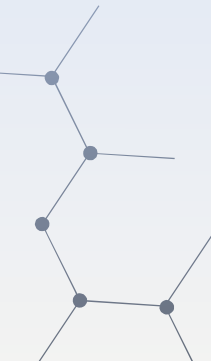
Session	Actions	Edit	View	Help
admin@ip-172-31-3-14:~\$	docker ps			
CONTAINER ID	NAME	COMMAND	CREATED	STATUS
89f028c271f	gchr.io/telium-security/adhoney:24.04.1	"/adhoney"	41 minutes ago	Up 41 minutes
74b6571c13c	gchr.io/telium-security/fatt:24.04.1	"/bin/sh -c 'python3..."	41 minutes ago	Up 41 minutes
c9cd96d000	gchr.io/telium-security/elasticsearch:24.04.1	"/bin/sh -c 'ARCH=\$(..."	41 minutes ago	Up 41 minutes (healthy)
0d42627e7fa	gchr.io/telium-security/compet:24.04.1	"/bin/sh -c 'exec ./u..."	41 minutes ago	Up 41 minutes (healthy)
93a187a8a72	gchr.io/telium-security/wodpost:24.04.1	"/bin/sh -c './usr/bin/..."	41 minutes ago	Up 41 minutes
19443a3a568	gchr.io/telium-security/redis:24.04.1	"redis-server /etc/ri..."	41 minutes ago	Up 41 minutes
630091f0a7f	gchr.io/telium-security/nap:24.04.1	"/bin/sh -c './usr/bin/..."	41 minutes ago	Up 41 minutes
74958caba72	gchr.io/telium-security/redpot:24.04.1	"/redpot"	41 minutes ago	Up 41 minutes
daed916d080	gchr.io/telium-security/compet:24.04.1	"/bin/sh -c 'exec ./u..."	41 minutes ago	Up 41 minutes (healthy)
53307d4ef23b	gchr.io/telium-security/phox:24.04.1	"python3 sandbox.py"	41 minutes ago	Up 41 minutes
4b021656215	gchr.io/telium-security/miniprint:24.04.1	"/server --bind 0.0.0.0..."	41 minutes ago	Up 41 minutes
23111307436	gchr.io/telium-security/cuerie:24.04.1	"/usr/bin/telnet -u..."	41 minutes ago	Up 41 minutes
5693897945d	gchr.io/telium-security/nginx:24.04.1	"nginx -g 'daemon of..."	41 minutes ago	Up 41 minutes
f431c550094	gchr.io/telium-security/honeytrap:24.04.1	"/opt/honeytrap/bin/..."	41 minutes ago	Up 41 minutes
5639764a4af	gchr.io/telium-security/dicompet:24.04.1	"/dicompet -ip 0.0.0.0..."	41 minutes ago	Up 41 minutes
3817d0928bc	gchr.io/telium-security/pdf:24.04.1	"/bin/sh -c 'exec ./u..."	41 minutes ago	Up 41 minutes
f4af92870ac	gchr.io/telium-security/redis:24.04.1	"redis-server /etc/ri..."	41 minutes ago	Up 41 minutes
0116a4479fb	gchr.io/telium-security/tptinit:24.04.1	"/opt/tpt/entrypoint..."	41 minutes ago	Up 41 minutes (healthy)

Limitation

- Advanced attackers may detect honeypots
- Controlled environment limitations
- Requires proper configuration

Future Research

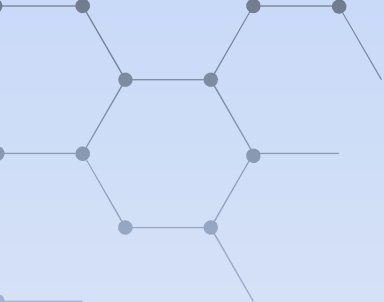
- Expand to larger networks
- Use high-interaction honeypots
- Apply machine learning
- Long-term analysis



Conclusion

- Honeypots are effective tools
- Capture real attacker behavior
- Provide actionable insights
- Support proactive security





References

- Alata et al. (2006)
- Pouget & Dacier (2011)
- Seifert et al. (2009)
- Franco et al. (2021)
- Deutsche Telekom Security (2024)

